CAAM/STAT31470: Homework #4

Purpose: The goal of these questions is to provide you practice for calculations and reinforce concepts from class. Unless specifically told in the question, you should always show your working and justify your answers. Marks will be awarded for your reasoning in addition to the correctness of your answers. Also note that these questions will be similar to the questions on the Midterm and Final.

Content: This week we look at contour integrals using the Cauchy-Goursat Theorem and its consequences, including the Cauchy Integral Formula. These tehniques can be very powerful, and already give us methods for computing interesting real integrals. We will also look briefly at the maximum modulus theorem.

Homework is due: **Wednesday May 7, 11:59pm**.

Problem 5.1 Midterm Revision

You have the opportunity to correct your solutions to the midterm exam for partial credit. The midterm exam questions can be found on the Modules page of canvas. Please submit any corrected midterm problems with this homework to gradescope. They will be graded and you will receive partial credit applied to your midterm grade.

For any question you submit here for regrading: clearly label which question or questions you are answering.

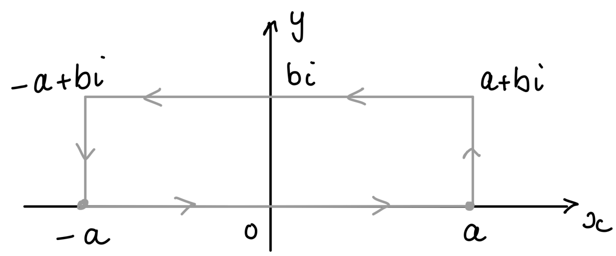
Problem 5.2 (12 points)

The goal of this question is to find the integral

$$\int_0^\infty e^{-x^2} \cos 2bx \, \mathrm{d}x = \frac{\sqrt{\pi}}{2} e^{-b^2} \tag{1}$$

for a real number $b > 0$

(a) Show that the integral of the function e^{-z^2} around the following rectangular contour is zero.



(b) Show that the sum of the integral over the lower and upper horizontal faces of the rectangle can be written as

$$2\int_0^a e^{-x^2} \, \mathrm{d}x - 2e^b \int_0^a e^{-x^2} \cos 2bx \, \mathrm{d}x$$

(c) Show that the integrals along the vertical faces of the rectangle can be written as

$$ie^{-a^2} \int_0^b e^{y^2} e^{-i2ay} \, \mathrm{d}y - ie^{-a^2} \int_0^b e^{y^2} e^{i2ay} \, \mathrm{d}y$$

(d) Hence, find an expression for the integral

$$\int_0^a e^{-x^2} \cos 2bx \, \mathrm{d}x$$

and take the limit as $a \to \infty$ to find the expression in equation (1).

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Problem 5.3 (6 points)

Let C be the positively oriented boundary of the half disk $0 \leq r \leq 1$, $0 \leq \theta \leq \pi$ and let $f(z)$ be defined by

$$f(z) = \begin{cases} \sqrt{r}e^{i\theta/2}, & r > 0 \\ 0 & r = 0 \end{cases}$$

By integrating over an appropriate parameterization of C, show that

$$\int_C f(z)\ \mathrm{d}z = 0$$

Why does the Cauchy-Goursat theorem not apply here?

Problem 5.4 (6 points)

Use Cauchy's integral formula to evaluate each of the following integrals where C is the square with side length 4 centered at the origin.

$$(a) \quad \int_C \frac{\cos(z)}{z(z^2+8)}\ \mathrm{d}z, \quad (b) \quad \int_C \frac{\cosh z}{z^4}\ \mathrm{d}z, \quad (c) \quad \int_C \frac{e^{iz}}{z(z^2+1)}\ \mathrm{d}z$$

Problem 5.5 (6 points)

Let f be continuous on a closed bounded region R, and let it be analytic and not constant throughout the interior of R. Assuming that $f(z) \neq 0$ anywhere in R, show that $|f(z)|$ has a minimum value m in R which occurs on the boundary of R and never in the interior.

Is the non-zero condition necessary for this result to hold?

End of homework